UNIT

IV

ENGINEERING PHYSICS • DIGITAL NOTES

Quantum Mechanics and Free Electron Theory

Matter waves • Uncertainty principle • Wave function • Schrödinger equations • Particle in a box • Classical and quantum free electron theory • Fermi–Dirac • Density of states • Fermi energy

QUICK SUMMARY • What you will learn

- Why particles behave like waves and what that means for what we can know about them
- How the Schrödinger equation gives quantised energies (particle in a box)
- How free electron theories explain conduction in metals, and Fermi–Dirac statistics, density of states and Fermi energy

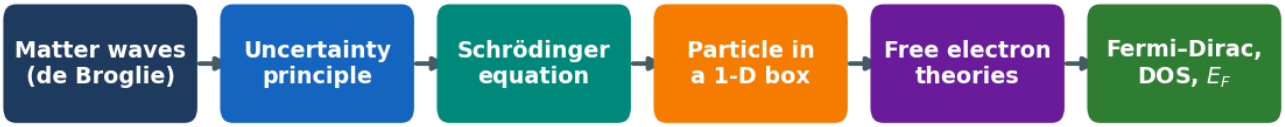


Fig. 1: Roadmap of Unit IV

1 Dual Nature of Matter

DEFINITION • de Broglie hypothesis (1924)

Every moving particle has an associated wave (matter wave) of wavelength λ = h/p, where p = mv is its momentum and h = 6.626 × 10^{-34} J·s.

de Broglie wavelength	$\lambda = \frac{h}{mv} = \frac{h}{p}$
Particle of kinetic energy E	$\lambda = \frac{h}{\sqrt{2mE}}$
Electron accelerated through V volts	$\lambda = \frac{h}{\sqrt{2meV}} = \frac{12.27}{\sqrt{V}}$ AA

EXAMPLE • Numerical

An electron accelerated through 100 V: λ = 12.27/√100 = **1.227 Å**, comparable to atomic spacing, so crystals can diffract electrons. A 0.15 kg cricket ball at 40 m/s has λ = 6.626×10^{-34}/6 = **1.1 × 10^{-34} m**, far too small to observe.

- **Davisson-Germer experiment (1927):** electrons of 54 V scattered from a nickel crystal show a strong peak at φ = 50°.
- Surface diffraction: d sinφ = nλ with d = 2.15 Å gives λ = 1.65 Å, matching the de Broglie value 12.27/√54 = 1.67 Å.
- This confirmed the wave nature of electrons (also G. P. Thomson's electron diffraction).

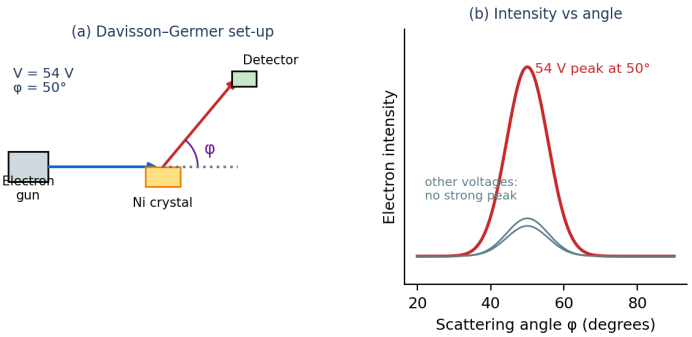


Fig. 2: Davisson–Germer electron diffraction

Properties of matter waves

- Associated with every moving particle, charged or neutral; λ decreases as mass or velocity increases.
- Not electromagnetic waves, they need no medium, and they cannot be observed for ordinary macroscopic bodies.

- Phase velocity $v_p = c^2/v$ (greater than c, no physical significance); group velocity $v_g = v$, equal to the particle velocity.
- The particle is represented by a **wave packet**, a group of waves with slightly different λ.

2 Heisenberg's Uncertainty Principle

DEFINITION • Statement
It is impossible to determine simultaneously and exactly both the position and the momentum of a particle. The product of the uncertainties is at least of the order of ħ/2.

Position–momentum	$\Delta x\,\Delta p \geq \frac{\hbar}{2} \qquad \left(\hbar = \frac{h}{2\pi}\right)$
Energy–time	$\Delta E\,\Delta t \geq \frac{\hbar}{2}$

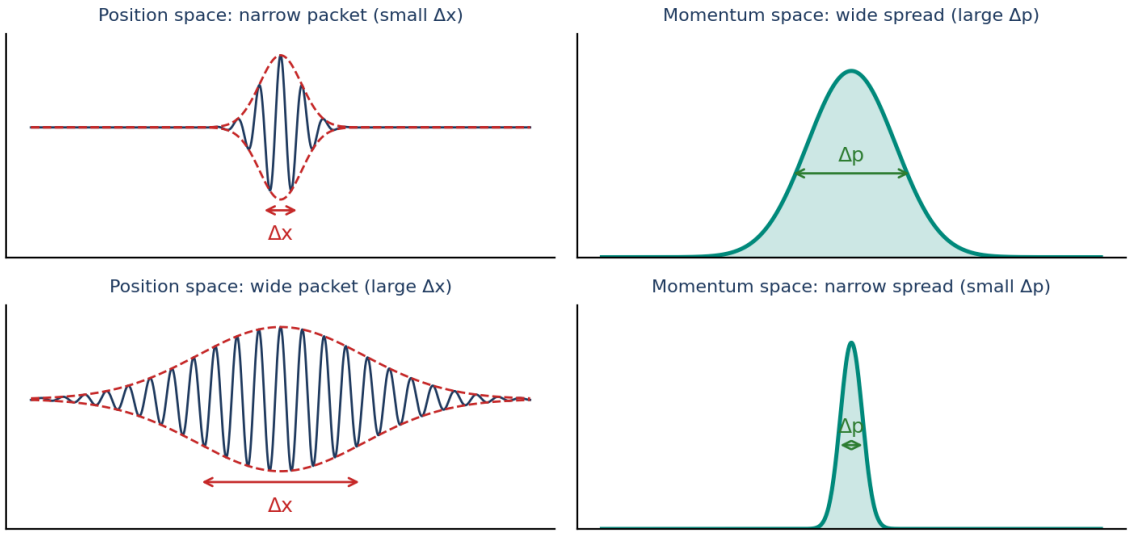


Fig. 3: A localised packet (small Δx) needs many wavelengths (large Δp), and vice versa

EXAMPLE • Can an electron live inside a nucleus?
Nuclear size $\Delta x \approx 10^{-14}$ m → $\Delta p \geq \hbar/2\Delta x = 5.3 \times 10^{-21}$ kg·m/s → $E \approx pc \approx 1.6 \times 10^{-12}$ J ≈ **10 MeV**. Beta electrons emerge with only 1–4 MeV, so electrons cannot exist inside the nucleus.

- **Significance:** it is a property of nature, not a limit of the instrument; it replaces exact orbits by probabilities.
- It explains the natural width of spectral lines (finite lifetime Δt) and why electrons have non-zero minimum (zero-point) energy in an atom or box.

3 Wave Function and Schrödinger Equations

DEFINITION • Wave function Ψ(x, t)
A complex function that describes the quantum state of a particle. It has no direct physical meaning, but $|\Psi|^2 = \Psi\Psi^*$ is the **probability density**: the probability of finding the particle in dx is $|\Psi|^2 dx$ (Born).

Property / condition	Meaning
Single-valued	One probability at every point
Finite and continuous	Probability cannot be infinite or jump
ψ and dψ/dx continuous	Needed for a finite second derivative (except at infinite potentials)
Normalisable	$\int \|\psi\|^2\, dx = 1$ over all space: the particle must be somewhere
Eigenfunction	Allowed solution for a given eigenvalue (energy)

Operators and expectation values (just the idea)

Observable	Operator	Eigenvalue equation
Position x	x (multiply)	—
Momentum p	$-i\hbar\, d/dx$	$p\psi = p\psi$
Energy (Hamiltonian H)	$-(\hbar^2/2m)\, d^2/dx^2 + V$	$H\psi = E\psi$

Probability in the interval (a, b)

$$P = \int_a^b |\psi|^2 dx$$

Average value of x

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi \, dx$$

Time-independent Schrödinger equation

Start with a matter wave of wavelength $\lambda = h/p$ and the classical wave equation for its amplitude: $d^2\psi/dx^2 + (4\pi^2/\lambda^2)\psi = 0$. Since $p^2 = 2m(E - V)$, $\lambda^2 = h^2/2m(E - V)$. Substituting:

1-D

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

3-D

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0 \iff H\psi = E\psi$$

Time-dependent Schrödinger equation

Write $\Psi(x, t) = \psi(x)\, e^{-iEt/\hbar}$. Differentiating with respect to t gives $i\hbar\, \partial\Psi/\partial t = E\Psi$; replacing $E\Psi$ using the time-independent equation:

Time-dependent equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi$$

REMEMBER • When to use which

The **time-independent** form (stationary states) applies when V does not depend on time, e.g. atoms, wells, solids. The **time-dependent** form is the general law of motion for Ψ.

4 Particle in a One-Dimensional Infinite Potential Well

A particle of mass m moves freely along x between rigid walls at x = 0 and x = L: V = 0 for 0 < x < L, V = ∞ elsewhere. Outside the well ψ = 0.



Fig. 4: Solution strategy

Inside the well: $d^2\psi/dx^2 + k^2\psi = 0$ with $k^2 = 2mE/\hbar^2$. General solution $\psi = A \sin kx + B \cos kx$.

- $\psi(0) = 0 \rightarrow B = 0$, so $\psi = A \sin kx$.
- $\psi(L) = 0 \rightarrow \sin kL = 0 \rightarrow kL = n\pi$, n = 1, 2, 3 … (n = 0 gives ψ = 0: no particle).
- Normalisation $\int_0^L A^2 \sin^2(n\pi x/L)\, dx = 1 \rightarrow A^2\, L/2 = 1 \rightarrow A = \sqrt{(2/L)}$.

Eigenfunctions

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\frac{n\pi x}{L}$$

Energy eigenvalues

$$E_n = \frac{n^2h^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

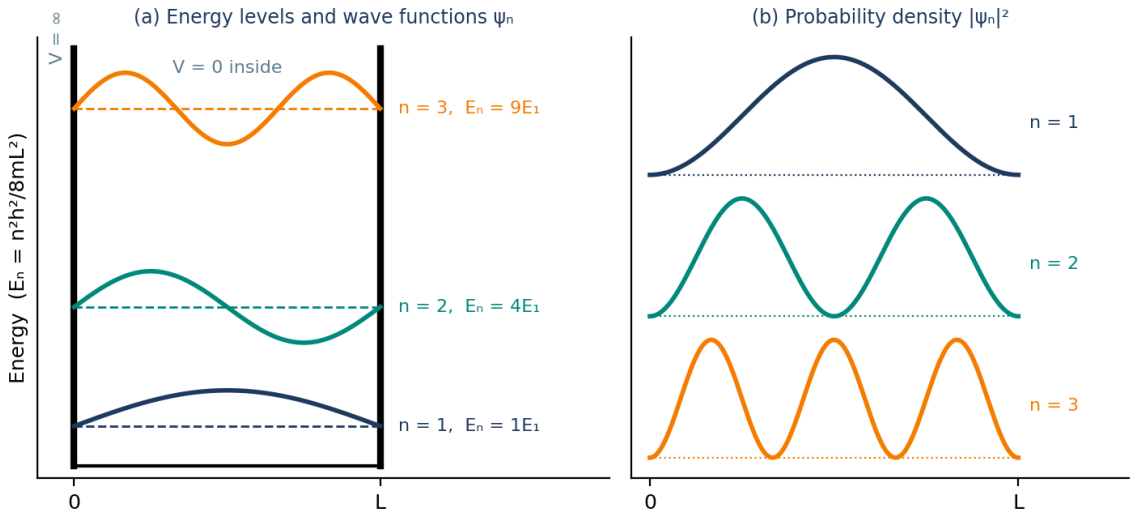


Fig. 5: Energy levels, wave functions and probability densities of a particle in a box

- **Energy is quantised:** only E_1, $4E_1$, $9E_1$ … allowed; spacing grows with n.
- **Zero-point energy** $E_1 = h^2/8mL^2 \neq 0$: the particle can never be at rest (uncertainty principle).
- n − 1 **nodes** inside the well. For n = 2 the probability of being at the centre is zero, while classically it is uniform.
- Smaller L → larger energies (quantum confinement, the basis of quantum dots). For large n the distribution becomes uniform (classical limit).

EXAMPLE • Numerical
Electron in a well of width L = 1 Å: $E_1 = h^2/8mL^2 = (6.626\times10^{-34})^2 / (8 \times 9.11\times10^{-31} \times 10^{-20}) = 6.0\times10^{-18}$ J = **37.6 eV**; $E_2 = 4E_1$ = **150.4 eV**. Photon for 2 → 1: 112.8 eV, λ = 12400/112.8 ≈ **110 Å**.

5 Classical Free Electron Theory (Drude-Lorentz)

Postulates: a metal contains a gas of free valence electrons moving randomly among fixed positive ions; electrons obey the **Maxwell–Boltzmann** classical statistics, and collisions with ions are the only interaction. An electric field adds a slow drift to the random motion.

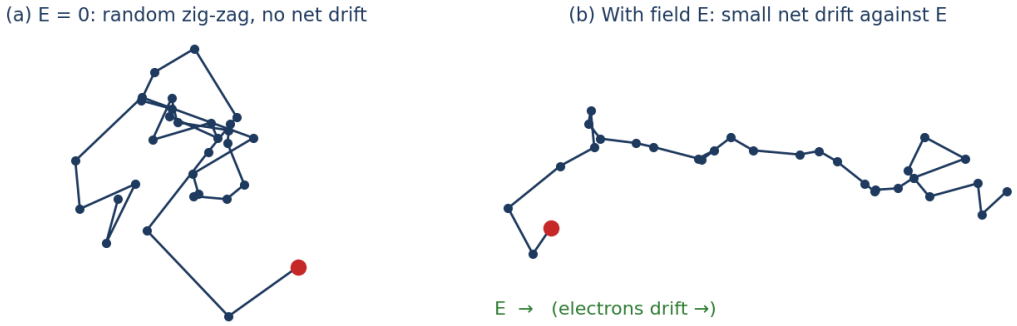


Fig. 6: Random motion of an electron with and without an applied field

Under field E the electron accelerates, a = −eE/m, until a collision. With mean free time (relaxation time) τ, drift velocity $v_d = a\tau$. Current density $J = n e v_d$:

Drift velocity and mobility	$v_d = \frac{eE\tau}{m}, \qquad \mu = \frac{v_d}{E} = \frac{e\tau}{m}$
Electrical conductivity (Ohm's law)	$\sigma = \frac{J}{E} = \frac{ne^2\tau}{m} = ne\mu$
Wiedemann–Franz law	$\frac{K}{\sigma T} = \frac{3}{2}\left(\frac{k}{e}\right)^2 = 1.12 \times 10^{-8}\ \mathrm{W\Omega K^{-2}}$

Merits	Demerits
Verifies Ohm's law	Fails to explain the electronic specific heat: predicts 3k/2 per electron, observed value is ~100 times smaller
Explains electrical and thermal conductivity	Predicts $\rho \propto \sqrt{T}$, but experimentally $\rho \propto T$
Gives the Wiedemann–Franz law (K/σT = constant)	Lorenz number 1.12×10^{-8} disagrees with the measured 2.44×10^{-8}
Explains metallic lustre and opacity	Cannot explain semiconductors, insulators, ferromagnetism, paramagnetism, photoelectric or Compton effect

6 Quantum Free Electron Theory (Sommerfeld)

Keeps the free electron gas picture but treats electrons **quantum mechanically**:

- Electrons move in a region of **constant potential** (inside the metal), a 3-D box, so their energies are quantised.
- Electrons obey the **Pauli exclusion principle**: at most two (spin up and down) per energy level, filled from the bottom up to the Fermi level.
- They follow **Fermi–Dirac statistics** instead of Maxwell–Boltzmann.
- Electrons are waves with λ = h/p; scattering happens only from **imperfections** and lattice vibrations, not from perfect ions.

Under field E the whole Fermi distribution shifts by $\delta k = -eE\tau/\hbar$, but only electrons near E_F can be scattered into empty states and contribute to the current. The result looks like the classical one with effective mass m*:

Conductivity (quantum)
$$\sigma = \frac{ne^2\tau}{m^*} = \frac{ne^2\lambda_F}{m^* v_F}$$

Mobility
$$\mu = \frac{e\tau}{m^*}$$

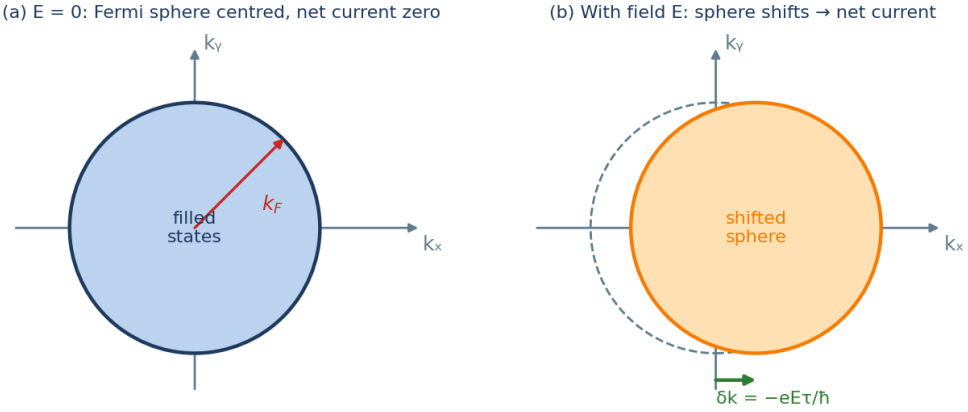


Fig. 7: Fermi sphere in k-space: the field shifts the whole sphere by δk

KEY IDEA • Why conductivity differs between metals
τ (or mean free path λ_F) depends on impurities and temperature; m* depends on how the band curves (Unit V). Resistivity rises with T because lattice vibrations reduce τ, giving $\rho \propto T$ as observed.

Merits of QFET	Demerits of QFET
Explains the small electronic specific heat and the T-independent paramagnetism	Assumes constant potential, ignoring the periodic lattice
Explains $\rho \propto T$ and gives the correct Lorenz number $(\pi^2/3)(k/e)^2 = 2.44\times10^{-8}$	Cannot explain why some solids are conductors, semiconductors or insulators (needs band theory)
Explains thermionic and photoelectric emission	Cannot explain positive Hall coefficients in some metals

7 Fermi–Dirac Distribution, Density of States and Fermi Energy

Fermi–Dirac distribution
$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

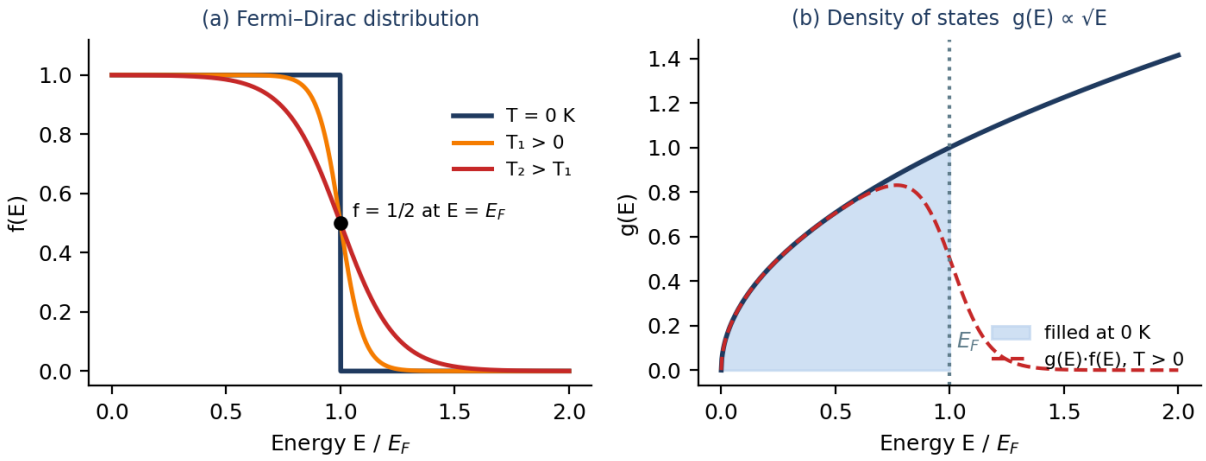


Fig. 8: Fermi–Dirac function and density of states in a metal

- f(E) is the **probability that a level of energy E is occupied** by an electron.
- **T = 0 K:** f = 1 for E < E_F and f = 0 for E > E_F (sharp step).
- **T > 0:** only electrons within ≈ kT of E_F are thermally excited; at E = E_F, f = 1/2 at any temperature.

DEFINITION • Fermi energy (E_F)
The energy of the highest occupied level at 0 K. At T > 0 it is the energy at which the occupation probability is 1/2. Typical values are 2–11 eV for metals (Cu 7.0 eV, Al 11.7 eV, Na 3.2 eV).

Density of states g(E)

g(E) dE is the number of electron states per unit volume between E and E + dE. In a cubic box of side L, $E = (h^2/8mL^2)(n_x^2 + n_y^2 + n_z^2) = n^2h^2/8mL^2$. In n-space the states with energy up to E fill the positive octant of a sphere of radius $n = (8mL^2E/h^2)^{1/2}$; with 2 spin states per level:

States up to E	$N(E) = 2 \times \frac{1}{8} \cdot \frac{4}{3}\pi n^3 = \frac{\pi}{3}\left(\frac{8mL^2E}{h^2}\right)^{3/2}$
Density of states (per unit volume)	$g(E)\,dE = \frac{4\pi}{h^3}(2m)^{3/2}E^{1/2}\,dE$

Fermi energy at 0 K

Electron concentration	$n = \int_0^{E_F} g(E)\,dE = \frac{8\pi}{3h^3}(2mE_F)^{3/2}$
Fermi energy	$E_F = \frac{h^2}{8m}\left(\frac{3n}{\pi}\right)^{2/3} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$
Mean energy at 0 K	$\bar{E} = \frac{3}{5}E_F$

EXAMPLE • Numerical
Copper: $n = 8.47 \times 10^{28}$ m^{-3}. $E_F = (\hbar^2/2m)(3\pi^2 n)^{2/3} = 6.10\times10^{-39} \times 1.84\times10^{20} = 1.13 \times 10^{-18}$ J = **7.03 eV**. Fermi temperature $T_F = E_F/k \approx 8.2 \times 10^4$ K, so at room temperature kT (0.026 eV) ≪ E_F and the electron gas is highly degenerate.

Classical versus quantum free electron theory

Aspect	Classical (Drude–Lorentz)	Quantum (Sommerfeld)
Statistics	Maxwell–Boltzmann	Fermi–Dirac (Pauli principle)
Energy levels	Continuous	Quantised, filled to E_F
Electrons contributing to heat and current	All free electrons	Only those within ~kT of E_F
Electronic specific heat	3k/2 per electron (too large)	$\propto T$, ~1% of classical (agrees with experiment)
Resistivity vs T	$\propto \sqrt{T}$ (wrong)	$\propto T$ (correct)
Electron nature	Particle	Wave, scattered only by imperfections

APPLICATIONS • Where this physics is used

- **Electron microscope:** fast electrons have $\lambda \approx 0.01$–1 Å, far below visible light, giving atomic resolution.
- **Quantum dots and wells:** $E_n \propto 1/L^2$, so size sets the emitted colour (displays, lasers, bio-imaging).
- **Tunnelling devices:** STM, tunnel diodes and flash memory use the leakage of ψ into barriers.
- **Metals and electronics:** E_F and g(E) determine heat capacity, thermionic emission and contact potentials.

REMEMBER • Quantum tunnelling (qualitative)

If a barrier is finite, ψ does not drop to zero at its wall; it decays exponentially inside and can emerge on the other side. A particle can therefore cross a barrier higher than its energy. This is impossible classically, and is used in tunnel diodes, STM and alpha decay.

8 Worked Problems

EXAMPLE • 1. de Broglie wavelength

Find the wavelength of a neutron ($m = 1.675\times10^{-27}$ kg) with kinetic energy 0.025 eV (thermal neutron). $E = 4.0\times10^{-21}$ J; $\lambda = h/\sqrt{(2mE)} = 6.626\times10^{-34}/\sqrt{(2 \times 1.675\times10^{-27} \times 4\times10^{-21})}$ = **1.81 Å**, suitable for diffraction by crystals.

EXAMPLE • 2. Uncertainty principle

The position of an electron is known to within 0.1 Å. $\Delta p \geq \hbar/2\Delta x = 1.055\times10^{-34}/(2 \times 10^{-11}) = 5.3\times10^{-24}$ kg·m/s, so $\Delta v \geq 5.8 \times 10^{6}$ m/s.

EXAMPLE • 3. Particle in a box

Electron in a 2 Å well: $E_1 = 37.6/4$ = **9.4 eV**; $E_3 = 9E_1$ = **84.6 eV**. The probability of finding the electron between x = 0 and L/4 in the ground state is $P = (2/L)\int \sin^2(\pi x/L)dx$ = **0.091**.

EXAMPLE • 4. Conductivity

Silver: $n = 5.86\times10^{28}$ m^{-3}, $\tau = 3.8\times10^{-14}$ s. $\sigma = ne^2\tau/m = 5.86\times10^{28} \times (1.6\times10^{-19})^2 \times 3.8\times10^{-14}/9.11\times10^{-31}$ = **6.3×10^{7} Ω^{-1}m^{-1}**; mobility $\mu = e\tau/m$ = **6.7×10^{-3} m²/Vs**.

EXAMPLE • 5. Fermi function

Probability of occupation at $E = E_F + 0.1$ eV at T = 300 K: $f = 1/(1 + e^{0.1/0.0259}) = 1/(1 + 47.5)$ = **0.0206** (about 2%). At $E = E_F - 0.1$ eV, f = **0.979**.

Key terms at a glance

Term	Meaning
Eigenvalue / eigenfunction	Allowed energy and its wave function, solution of $H\psi = E\psi$
Degenerate states	Different states having the same energy
Relaxation time τ	Average time between two successive collisions of an electron
Drift velocity	Average velocity gained along the field direction
Fermi level / Fermi surface	Highest filled level at 0 K / its boundary in k-space
Effective mass m*	Mass an electron appears to have in a crystal, reflecting the periodic potential

Model answers to common 2-mark questions

KEY IDEA • Why can't we see the wave nature of a cricket ball?
Its momentum is large, so $\lambda = h/p \sim 10^{-34}$ m, far smaller than anything that could act as a diffracting object.

KEY IDEA • Why is the ground-state energy of a particle in a box not zero?
If E = 0 then ψ = 0 everywhere (no particle). Confining the particle to length L makes $\Delta p \geq \hbar/2L$, so it must have kinetic energy of at least about $h^2/8mL^2$.

KEY IDEA • Why does only a small fraction of electrons conduct in a metal?
Pauli exclusion fills all lower states, so only electrons within ~kT of E_F have empty states to move into.

KEY IDEA • What happens to E_F when temperature rises?
For metals it changes very little, because $kT \ll E_F$; only the edge of the distribution is blurred.

EXAM CORNER • Try yourself (answers in brackets)

- Find the de Broglie wavelength of an electron accelerated through 400 V. [0.61 Å]
- A proton and an electron have the same kinetic energy. Which has the shorter wavelength, and by what ratio? [proton; $\sqrt{1836} \approx 43$ times shorter]
- Find E_1 and E_2 for an electron in a 5 Å box. [1.5 eV, 6.0 eV]
- Calculate E_F for aluminium ($n = 18.1 \times 10^{28}$ m^{-3}). [≈ 11.7 eV]
- The lifetime of an excited state is 10^{-8} s. Estimate the minimum energy uncertainty. [$\approx 3.3 \times 10^{-8}$ eV]

9 Formula Sheet and Exam Corner

Topic	Key formulae
de Broglie	$\lambda = h/p = h/\sqrt{(2mE)}$; electrons: $\lambda = 12.27/\sqrt{V}$ Å
Uncertainty	$\Delta x \Delta p \geq \hbar/2$; $\Delta E \Delta t \geq \hbar/2$
Schrödinger	$d^2\psi/dx^2 + (2m/\hbar^2)(E - V)\psi = 0$; $i\hbar\partial\Psi/\partial t = H\Psi$
Particle in box	$\psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$; $E_n = n^2h^2/8mL^2$
Classical FET	$\sigma = ne^2\tau/m$; $\mu = e\tau/m$; $K/\sigma T = (3/2)(k/e)^2$
Quantum FET	$\sigma = ne^2\tau/m^*$
Fermi	$f(E) = 1/[1 + e^{(E-E_F)/kT}]$; $g(E) = (4\pi/h^3)(2m)^{3/2}E^{1/2}$; $E_F = (h^2/8m)(3n/\pi)^{2/3}$

EXAM CORNER • Short questions (2–3 marks)

- State de Broglie's hypothesis. Derive λ for an electron accelerated through V volts.
- State Heisenberg's uncertainty principle.
- What is the physical significance of the wave function?
- Write the conditions a wave function must satisfy.
- Write the Schrödinger time-independent equation.
- Define Fermi energy and Fermi level.
- State the Wiedemann–Franz law.

EXAM CORNER • Long questions (10 marks)

- Derive the Schrödinger time-independent and time-dependent wave equations.
- Solve Schrödinger's equation for a particle in a 1-D infinite potential well and obtain energy eigenvalues and eigenfunctions.
- Explain the classical free electron theory. Derive an expression for electrical conductivity and discuss its merits and demerits.
- Discuss quantum free electron theory and derive $\sigma = ne^2\tau/m^*$.
- Explain the Fermi–Dirac distribution function and its variation with temperature.
- Derive the density of states and hence the Fermi energy at 0 K.

EXAM CORNER • Quick MCQs

- 1. Zero-point energy of a particle in a box is (a) 0 (b) $h^2/8mL^2$ (c) $h^2/2mL^2$.
- 2. At $E = E_F$, f(E) at $T > 0$ is (a) 0 (b) 1/2 (c) 1.
- 3. g(E) is proportional to (a) E (b) $\sqrt{E}$ (c) 1/E.
- 4. Classical theory predicts resistivity $\propto$ (a) T (b) $\sqrt{T}$ (c) 1/T.
- **Answers:** 1-b, 2-b, 3-b, 4-b.

QUICK SUMMARY

- Matter has wave nature ($\lambda = h/p$); $\Delta x \Delta p \geq \hbar/2$.
- Schrödinger equation in a box gives $E_n = n^2h^2/8mL^2$ and $\psi_n = \sqrt{(2/L)} \sin(n\pi x/L)$.
- Classical theory (Maxwell–Boltzmann) gets Ohm's law but fails for specific heat; quantum theory (Fermi–Dirac, Pauli) fixes this: $g(E) \propto \sqrt{E}$, $E_F = (h^2/8m)(3n/\pi)^{2/3}$.